

Web Application Development Practical - Docker Container Environment

Aim: To create Docker Container Environment and host a web application inside Docker container.

Theory:

Docker is a containerization platform used to package applications with all dependencies into lightweight containers. Containers provide portability, fast deployment and isolation. Docker images are used to create containers and Docker Engine manages the execution of containers.

Requirements:

1. Docker Desktop Installed
2. VS Code or Terminal
3. Internet Connection
4. Basic HTML Knowledge

Detailed Procedure

Step 1: Download and install Docker Desktop.

Step 2: Verify Docker installation using `docker --version` command.

Step 3: Create project folder named `docker-project`.

Step 4: Create `index.html` file inside project folder.

Step 5: Create `Dockerfile` inside same directory.

Step 6: Write Docker instructions in `Dockerfile`.

Step 7: Open terminal in project directory.

Step 8: Build Docker image using `docker build` command.

Step 9: Verify image creation using `docker images` command.

Step 10: Run container using `docker run` command.

Step 11: Open browser and visit `localhost:8080`.

Step 12: Check running containers using `docker ps` command.

Project Structure

```
docker-project/  
├── Dockerfile  
└── index.html
```

index.html

```
<!DOCTYPE html>  
<html>  
<head>  
  <title>Docker Environment</title>  
  <style>
```

```

        body{
            font-family:Arial;
            background:#f4f4f4;
            text-align:center;
            padding:50px;
        }

        .container{
            background:white;
            padding:30px;
            border-radius:10px;
            box-shadow:0px 2px 8px gray;
        }

        h1{
            color:#007bff;
        }
    </style>

</head>

<body>

<div class="container">

    <h1>Docker Container Environment</h1>

    <p>Website Hosted Successfully using Docker Container</p>

</div>

</body>
</html>

```

Dockerfile

```

FROM nginx:latest

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80

```

Docker Commands

```

docker --version

docker build -t mydockerapp .

docker images

docker run -d -p 8080:80 mydockerapp

docker ps

docker stop <container_id>

```

Advantages of Docker

1. Lightweight and Fast
2. Platform Independent
3. Easy Deployment
4. Better Resource Utilization
5. Simplifies Application Management

Explanation:

Docker containers package applications with dependencies into isolated environments. The Dockerfile contains instructions to create Docker image. Nginx web server is used to serve the HTML webpage inside the container. The application becomes portable and can run consistently on different systems.

Result: Docker Container Environment was created successfully and the web application was hosted inside Docker container.